

Words and Money at ECOSOC

Do the UN's stated priorities match where its money flows?

Tahvia Williams, Ada Meltzer, Vadim Rudic, Enrico Madani

BUSN 20800: Big Data, Spring 2026 | May 26, 2026

Abstract

We ask whether the topics the UN Economic and Social Council (ECOSOC) discusses align with where UN money goes, and whether ECOSOC text helps predict that money. We collect 910 ECOSOC summary records (2000 to 2025) and a 244-row panel of UN agency revenue (2011 to 2024), then build feature sets of increasing richness (activity and macro controls, theme word counts, and TF-IDF). We map the agenda with an LDA topic model checked against hand labels (Q1), and we predict year-over-year UN revenue growth with five models across nested feature sets using a time-ordered train, validation, and test split (Q2). We find that ECOSOC talk is heavily focused on one development theme while the money it coordinates is spread across many agencies, so the agenda and the money point the same way but do not match in detail. Adding text features lowers out-of-sample error for the linear models, and the best result comes from OLS with TF-IDF (test RMSE 0.094). The main limit is the small yearly sample, which we handle with a rolling backtest and by trusting results that hold across feature sets.

1. Introduction

Large institutions say a lot about what they care about. They give speeches, pass resolutions, and publish meeting records. It is much harder to see whether that talk matches where the money goes. This paper asks a plain version of that question for the UN Economic and Social Council (ECOSOC): when ECOSOC talks about a topic, does the funding follow?

ECOSOC is a good place to test this. It has a clear job coordinating the UN's economic, social, and environmental work, and it publishes meeting records that go back decades, so we have a long run of text to read. It also sits right next to the money: the UN agencies that report to it raise and spend tens of billions of dollars each year. That lets us line up what is said in a given year against what is funded in the same year.

We focus on two questions:

1. **Q1 (Alignment).** Do ECOSOC themes line up with UN funding patterns?
2. **Q2 (Prediction).** Do richer text features help predict UN revenue growth?

Our main finding is that the agenda and the money agree in direction but not in detail. ECOSOC talk is dominated by one broad development theme in almost every year, while the money is split across food, health, refugees, and children. Text features do help predict next-year revenue growth, most clearly in the linear models, but the gains are model-dependent and need to be read with care given how few years of data we have.

2. Data

We use three datasets. All the code that collects and cleans them is in the project repository; we do not submit any raw data, as the assignment asks.

2.1 ECOSOC summary records

We collected 910 ECOSOC meeting summary records from 2000 to 2025 from the UN Digital Library. Each record is a PDF that starts with a procedural header (who attended, the agenda, the room and time) and then summarizes what was said and decided. We clean out the header text, then group the text by year so it lines up with the yearly money data. Coverage is thin in 2000 and builds up through the 2000s.

2.2 UN agency revenue

From the UN System Chief Executives Board (CEB) financial statistics we built a 244-row table covering 19 ECOSOC-related agencies from 2011 to 2024. For each agency and year we have its revenue. We also add up all agencies in a year to get total UN revenue, which is the thing we try to predict in Q2.

2.3 Macro controls (WDI)

The macro control variables come from the World Bank Development Indicators (WDI) dataset, which covers 266 countries from 2000 to 2024. We pulled seven indicators: GDP growth, government expenditure as a share of GDP, life expectancy, official development assistance received, the GINI coefficient, FDI inflows, and secondary school enrollment. Of these, four (GDP growth, government expenditure as a share of GDP, life expectancy, and ODA received) are used as controls in the supervised baseline. They stand in for broad global economic and development conditions that could move UN revenue on their own, so any gain from text features is measured on top of them.

2.4 Summary table

Table 1 lists the row counts, years covered, and key variables for each source. The text corpus and the revenue panel are both grouped to the year level for the prediction task, which leaves a short yearly series. That short series is the reason for the sample-size caution in Section 5.

Data source	Rows	Years	Note
ECOSOC records	910 records	2000-2025	text corpus
UN agency revenue	244 agency-years	2011-2024	19 agencies
WDI controls	7 indicators	2000-2024	266 countries

Table 1. Data sources used in the project.

3. Methods

3.1 Text preprocessing

We start by stripping the standard ECOSOC header from each record (room number, attendance list, meeting open time), because that text is the same in every meeting and would otherwise drown out the real content. We then split the text into sentences and group everything by year, which gives us 26 year-level documents. Working at the year level keeps the text lined up with the yearly money data.

3.2 Feature tiers used in the final supervised scope

We build three feature sets that get richer in steps, so we can see how much each layer of text adds:

- **Numeric_only:** basic activity and sentiment features plus the macro controls.
- **Plus Tier 1:** adds theme word-count rates (how often development, health, gender, and other theme words appear).
- **Plus Tier 2:** adds TF-IDF features, which capture specific one- and two-word phrases.

3.3 Unsupervised topic model

To describe the agenda, we fit an LDA topic model to the yearly text. LDA finds groups of words that tend to appear together and calls each group a topic. We tried 6, 10, and 15 topics and picked the number that scored best on a standard coherence measure, which was 6. We then read the top words of each topic and gave it a plain theme label (development, health, climate, and so on). To check the topics are real and not just leftover procedural language, we hand-labeled a 52-record sample by year and measured how often the human label showed up in the model's top three themes for that year.

Agreement was about 60%. It was near-perfect for development and health and zero for humanitarian, gender, and conflict, which makes sense because at 6 topics the model only separates a few themes.

3.4 Supervised setup

Target Variable: We estimate whether ECOSOC text features improve out-of-sample prediction of annual UN revenue growth. Let UN_t denote total UN revenue in year t , the dependent variable is

defined as: $y_t = \log(\frac{UN_t}{UN_{t-1}})$

Thus, model coefficients and prediction errors are interpreted on a year-over-year growth-rate scale.

Model design: To preserve temporal ordering and avoid information leakage, we use a chronological split: training years 2012-2019, validation years 2020-2021, and test years 2022-2024. We compare five supervised regressors: OLS, Ridge, Lasso, Random Forest, and Gradient Boosting. This model set spans unregularized linear, regularized linear, and non-linear tree-based learners. This allows us to test whether any observed gains are model-specific.

We evaluate three nested feature sets:

1. numeric_only: meeting/activity features plus macro controls;
2. +tier1: numeric_only plus lexicon-based thematic rates;
3. +tier2: +tier1 plus TF-IDF text features.

To minimize OVB, we include macro controls from WDI, as listed in the Data section above, in the baseline feature block. These variables proxy broad global economic and development conditions that may influence UN revenues independently of ECOSOC language. Any performance improvement from +tier1 or +tier2 is measured relative to a macro-adjusted baseline.

OLS is the unregularized linear baseline; Ridge ($\alpha = 1.0$) adds moderate L2 shrinkage to stabilize high-dimensional coefficients; Lasso ($\alpha = 0.01$, $\text{max_iter} = 20000$) adds L1 shrinkage to encourage sparsity and interpretability. For tree-based models, we use RF ($\text{n_estimators} = 200$, $\text{max_depth} = 3$, $\text{random_state} = 42$) and GBM ($\text{n_estimators} = 200$, $\text{max_depth} = 3$, $\text{random_state} = 42$) with shallow trees to limit overfitting in a small annual sample while preserving non-linear capacity.

Linear models are fit in a 'StandardScaler' pipeline so regularization penalties are comparable across differently scaled predictors. Tree models are left unscaled because split-based methods are scale-invariant.

4. Results

4.1 Q1: Alignment

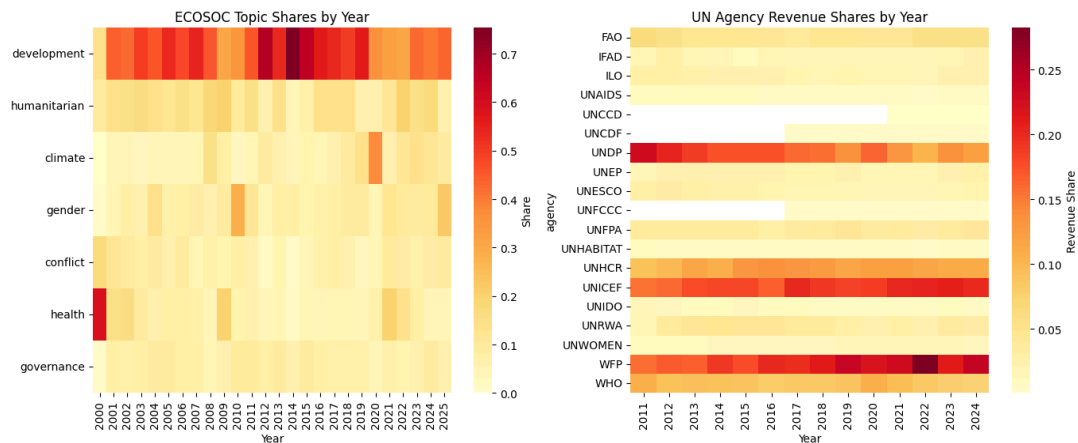


Figure 1. ECOSOC Topic Shares and UN Agency Revenue Shares by Year

Figure 1 compares ECOSOC's agenda with the UN's agency revenue share from 2011 to 2024. Across the whole period, development dominates ECOSOC's discussions, reaching up to 70% of the topic share in 2014. This lines up with the revenue lead of UNDP and UNICEF, which are both development-focused agencies. WFP's revenue grew sharply after 2020, reaching about 25% of the revenue share in 2022, likely driven by COVID-19 food insecurity and the war in Ukraine. But there is no matching jump in humanitarian language in ECOSOC meetings, which suggests the talk lags the funding. We also see a spike in health language in 2000, when the AIDS crisis put global health at the center of the debate.

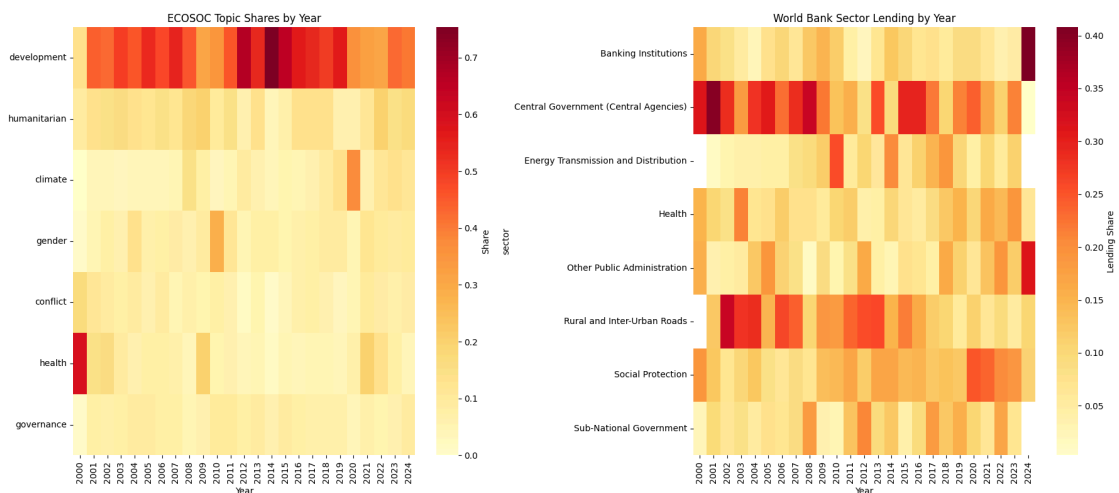


Figure 2. ECOSOC Topic Shares and World Bank Sector Lending Shares by Year

Figure 2 compares ECOSOC's agenda with World Bank sector lending from 2000 to 2024. Even though development and health dominate ECOSOC talk, World Bank lending stays concentrated in Central Government, which reaches up to 40% of lending in 2001, and in Rural and Inter-Urban Roads. Health lending peaks around 2002 to 2004 and again around 2021 to 2023, which lines up with periods of more health talk in ECOSOC. Banking Institutions lending jumps in 2024 while Central Government lending falls.

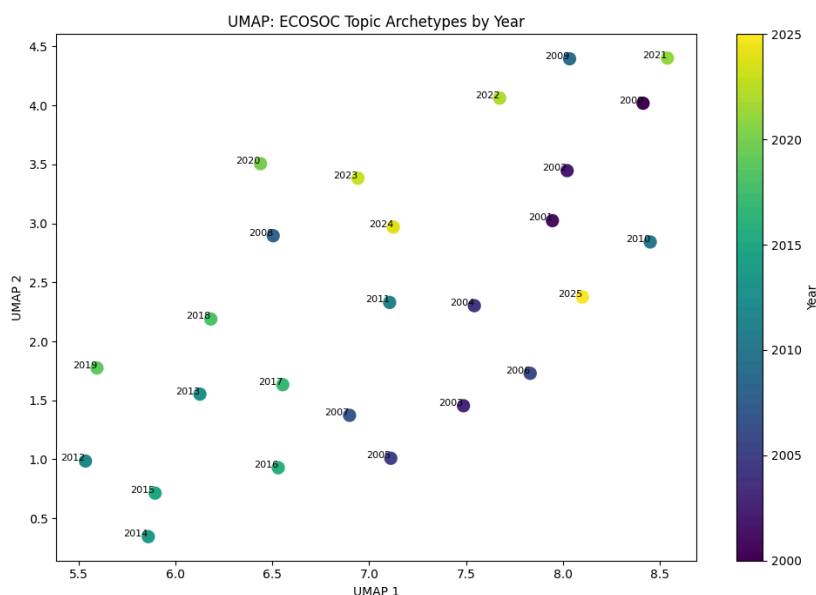


Figure 3. UMAP projection of ECOSOC yearly topic mixtures

Figure 3 is a UMAP projection of ECOSOC's yearly topic mixtures. Each point is a year, and points that sit close together had similar topic patterns. The year 2000 sits alone at the top, which reflects its health-heavy agenda during the AIDS crisis. The early 2000s cluster on the right, the mid-2010s cluster in the bottom left, and recent years from 2019 on are more scattered. That scatter suggests ECOSOC's agenda has become more varied and harder to predict lately.

4.2 Q2: Prediction

We use RMSE as the main metric because it measures out-of-sample error directly in the target's units (log-growth), which is easy to read. We do not lean on R-squared here because it is unstable in our small test set and often goes far negative.

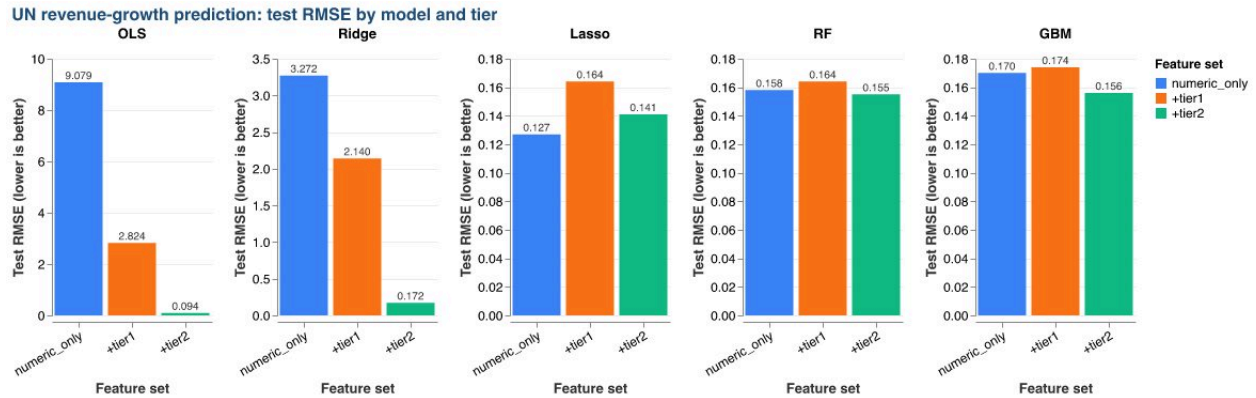


Figure 4. Test RMSE by model and feature tier. Note the model-specific vertical scales. Text features help the linear models most.

Figure 4 reports test RMSE by model and feature tier. The lowest fixed-split error is OLS plus Tier 2 (RMSE 0.094). OLS drops sharply as the text gets richer (9.079 numeric-only, 2.824 plus Tier 1, 0.094 plus Tier 2), and Ridge follows the same path (3.272, 2.140, 0.172). The tree models improve only a little with Tier 2 (Random Forest 0.158 to 0.155; Gradient Boosting 0.170 to 0.156). Lasso is mixed, doing best with numeric-only (0.127) and slightly worse once text is added. Lasso remained sparse, selecting only a few predictors: `gdp_growth` and `oda_received` in `numeric_only`; `gdp_growth` plus `lex_humanitarian`, `lex_development`, and `lex_gender` in `+tier1`; and `gdp_growth` plus four TF-IDF terms in `+tier2`.

Overall, richer text, especially the Tier 2 TF-IDF features, tends to improve prediction, though the size of the gain depends on the model. The strongest gains show up in the linear models, while the tree models change less but mostly in the same direction. Given the small yearly sample, we read these results as pointing in a clear direction rather than as exact, settled effects.

5. Discussion + Limitations

A few cautions shape how far these results should be pushed.

- **Small sample:** We have only a handful of years of revenue data against thousands of possible text features. That makes in-sample fit easy to inflate and test scores jumpy, which is exactly why the numeric-only linear models post huge errors and the Tier 2 models post very small ones. The rolling backtest and our preference for results that hold across models are our main defenses.
- **Summary text, not transcripts:** ECOSOC publishes summaries, not word-for-word transcripts, so some of what was actually said is smoothed over or dropped. We filter the boilerplate and validate the topics, but the text is still a summary.

- **Association, not cause:** Our models show patterns, not causes. The macro controls reduce confounding from broad global trends, but we do not claim that ECOSOC words cause funding to change.
- **Why controls matter:** Without the macro controls, global booms or downturns could look like a text effect. Including them means any text gain is measured on top of those broad conditions.

A natural next step is to work at the meeting level instead of the year level, which would turn 26 yearly points into hundreds of observations and make the prediction results much steadier.

6. Conclusion

On Q1, ECOSOC's words and the money it coordinates point the same way but do not match in detail: the agenda is dominated by one development theme which consistently occupied over 50% of the topic share throughout the period, while the money is spread across many agencies. On Q2, ECOSOC text does carry real signal about next-year UN revenue growth, strongest in the linear models with TF-IDF features (OLS test RMSE 0.094), though the small sample means we read it as directional. The practical takeaway is that stated priorities and spending rhyme without lining up exactly, and the clearest next step is to push the analysis to the meeting level so the patterns can be pinned down with more data.

References

- UN Digital Library. ECOSOC summary records (E/...). documents.un.org.
- UN System Chief Executives Board for Coordination (CEB). Financial Statistics. unsceb.org.
- World Bank. World Development Indicators (WDI).
- Tools: gensim (LDA), scikit-learn (supervised models), sentence-transformers, VADER.

Appendix A. Member contributions

Member	Contribution
Tahvia Williams	Unsupervised topic modeling (topic_model.ipynb) and topic validation (label_transcripts.ipynb), plus the topic interpretation text. Wrote Sections 3.1, 3.2, 3.3, and the Q1 narrative in 4.1.
Vadim Rudic	Data controls and alignment prep: the WDI controls and the alignment visuals (the two heatmap comparisons and the UMAP), and support on the Q1 interpretation with Tahvia. Wrote Section 2.3, 4.1 the Conclusion.
Ada Meltzer	Corpus and data summary statistics, Table 1, descriptive QA checks, and formatting support. Drafted the Abstract and Introduction, and led Section 2.
Enrico Madani	Supervised pipeline (models.ipynb): the growth target, time-ordered split, the five models across feature sets, ablations, robustness checks, and supervised figures. Wrote Sections 3.4, 4.2, and 5.